

3rd Progress Report

W. Chan

Date:	31/1/2024
Subject:	RL for Flying-V v2.0
Supervisors:	Dr. E.J. van Kampen
Project term:	01/2024 - 08/2024
E-mail:	w.y.chan@student.tudelft.nl

Outcome of the previous meeting

- Got names of other people working on flying-v control, and of people who left C&S and brought incremental dynamic programming with them
- Reviewed latest draft of research questions, several questions need to be made more specific, and mindful of the amount of work needed to answer each question.
- Reviewed potential avenues for the thesis project:
 1. Attitude control of flying-v (poor lateral flight qualities in flying wing designs).
 2. Online-offline hybrid with novelty in either the offline or online components (offline using Marek's RUN-DSAC, online using Zhou's IGDHP).
- Got advice on the literature study and some assurance that the literature review is going as it should.

Description of the results since last meeting

- Identified two avenues to applying RL to AFCS that would be interesting to explore:
 1. **RL gain tuning**¹: Using RL algorithms to provide control gains instead of control actions. Instead of the action space being control surface deflections or deflection increments, the action space could instead be made to be gains of a PID controller, or similarly the gain increments of a PID controller.
 2. **Multistep IDHP**²: Extending the IDHP algorithm to a multi-step version, instead of the existing algorithm where only the previous immediate reward is used to update the estimate of the value function gradient. Alternatively, IDHP can be augmented with eligibility traces instead of the n-step approach, which as far as I could tell gives the same ultimate effect while being computationally more efficient.

¹Inspired by this paper which used DDPG to learn an agent that can provide gains to a controller online: <https://doi.org/10.1007/s11071-023-08444-4>

²Inspired by this paper which developed a multistep HDP algorithm: <https://doi.org/10.1016/j.ins.2017.05.005>

- Further iterated on research questions:

Research Objective:

To advance the state-of-the-art reinforcement learning fault tolerant flight controllers and further the technological readiness level of the Flying-V, by developing a reinforcement learning based intelligent flight control system that can stabilize and improve the handling qualities of the Flying-V.

Research Questions:

1. What reinforcement learning algorithm can yield a flight controller that is the most tolerant to faults while providing the best reference tracking performance??
 - (a) What RL algorithms are considered to be state-of-the-art?
 - (b) In what ways are RL-based flight controllers fault-tolerant, and how can they be made as fault-tolerant as possible?
 - (c) Which algorithm demonstrates the highest reference tracking accuracy and fault tolerance when applied to a system with some simple dynamics?
2. What are the flight control related challenges when it comes to designing a controller for the Flying-V?
 - (a) What are potential fault scenarios that warrant attention from the AFCS system in the Flying-V?
 - (b) What are the flight handling qualities of the Flying-V and how can they be assessed when augmented with an RL-based AFCS?
3. How can the identified algorithm be applied to control the Flying-V?
 - (a) How should the flight control system be structured?
 - (b) What is the MDP in the case of controlling the Flying-V, what are the states, observations, actions, and environment?
4. How does the implemented flight controller perform both in the presence of faults and nominal flight?
 - (a) What criteria should be kept track of to characterize a controller's fault tolerance (fault tolerance success rate, nMAE of tracking error, variance of nMAE...)?
 - (b) Noting the possible fault scenarios and handling qualities of the Flying-V, what flight scenarios should be designed to test the performance of the controller?
 - (c) How well does the nominal flight performance of the proposed flight controller compare to other state-of-the-art controllers?
 - (d) How are the fault tolerance characteristics of the proposed flight controller compared to other state-of-the-art controllers?

Plans for next 2 weeks

1. I will take a spring holiday in Slovenia for 1 week from the 4th of Feb.
2. More reading, more reading, more reading! Continue gathering and studying literature from the field of reinforcement learning but with more emphasis on the intersection between reinforcement learning and flight control.
3. Possibly start freezing the research questions.